25/04

Reference variable

Reference variable is an alias name given to the existing variab

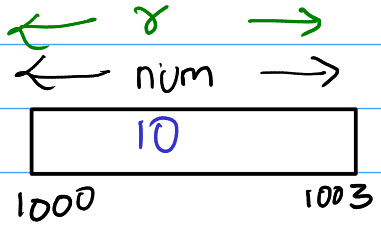
syntax :

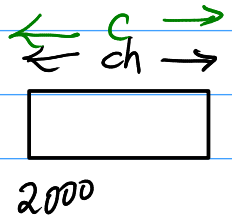
datatype & refname=existing variable

ex:

```
int num=10;
int &r=num; // r is reference to num

char ch;
char &c=ch; // c is reference to ch
```





reference variable does not occupy the memory, it refers to existing variable memory block itself.

Once reference is tieup with variable througout its lifetime it refers only to it.

We can create references only to existing variables.

We can create more than one reference to existing variables.

We cannot create reference to an array

References must be iniatialised only in the declaration.

ex : int num=10;

```
int &r; //error
r=num;
```

or

```
&r=num; //error
```

↳ it is acting as address operator

We cannot create array of references

We can create reference to a pointer

syntax :

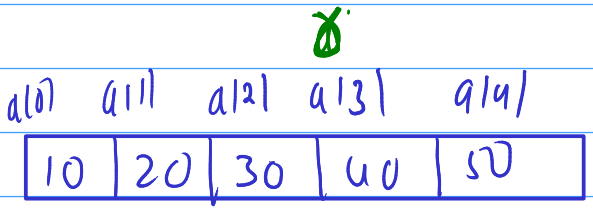
datatype * &ref=pointer;

ex : int *p;

int * &q=p;

Reference cannot be NULL.
Reference cannot be generic or void type.

```
int a[5]={10,20,30,40,50};
int &r=a[3];
```



syntax to create a reference to constant

```
const datatype &ref=const;
```

ex : const int &ref=10;
const int x=34;
const int &r=x;

How do we pass values to functions?
In C we have two methods
1. call by value
2. call b y reference

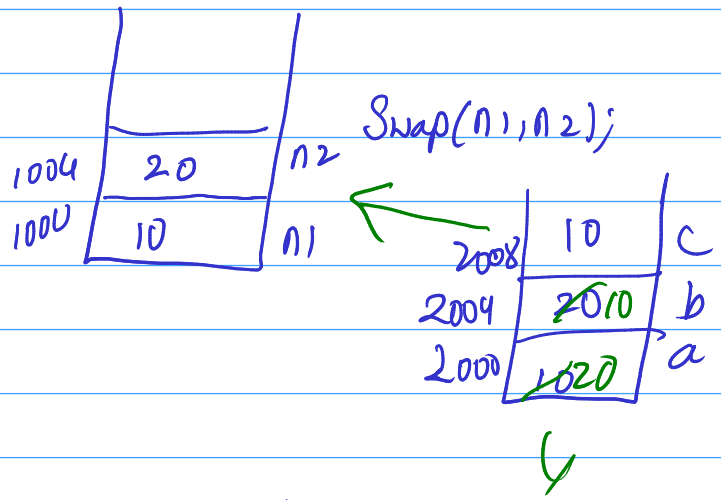
In C++ we have 3 methods
1. call by value
2. call by address(same as call by reference in C)
3. call by reference

In C++ call byb reference , formal arguments are references to actual arguments.

void Swap(int a=n1, int b=n2)

```
void Swap(int a,int b)
{int c=a;
a=b;
b=c;
}

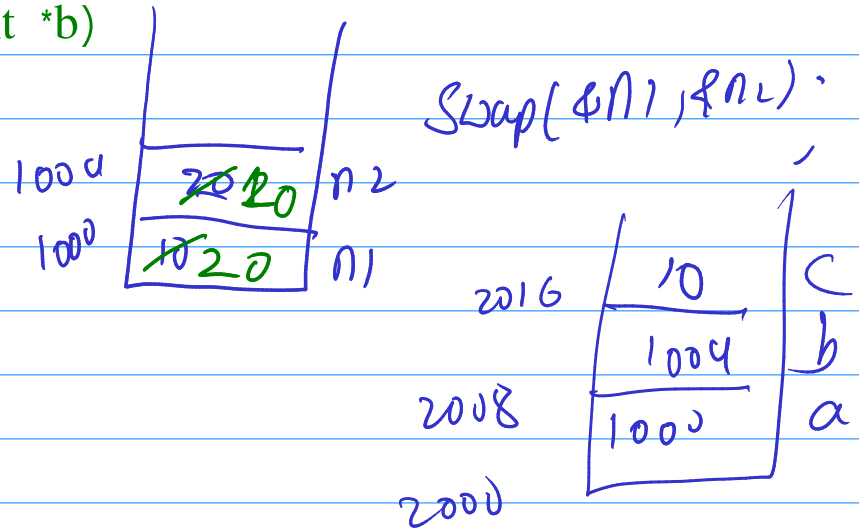
int main()
{int n1=10,n2=20;
Swap(n1,n2);
}
```



void Swap(int *a=&n1, int *b=&n2)

```
void Swap(int *a,int *b)
{ int c=*a;
*a=*b;
*b=c;
}

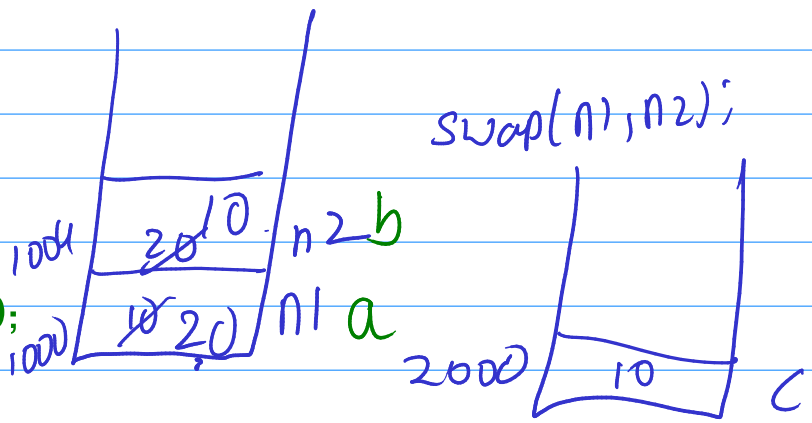
int main()
{ int n1=10,n2=20;
Swap(&n1,&n2);
}
```



void Swap(int &a=n1, int &b=n2)

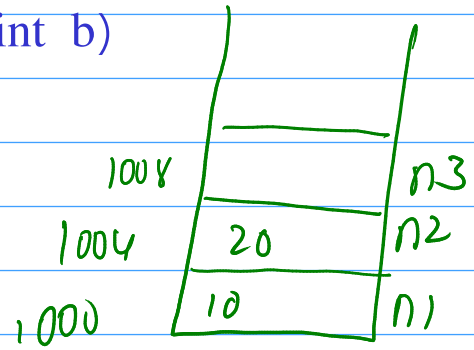
```
void Swap(int &a,int &b)
{ int c;
        c=a;
        a=b;
        b=c;
}

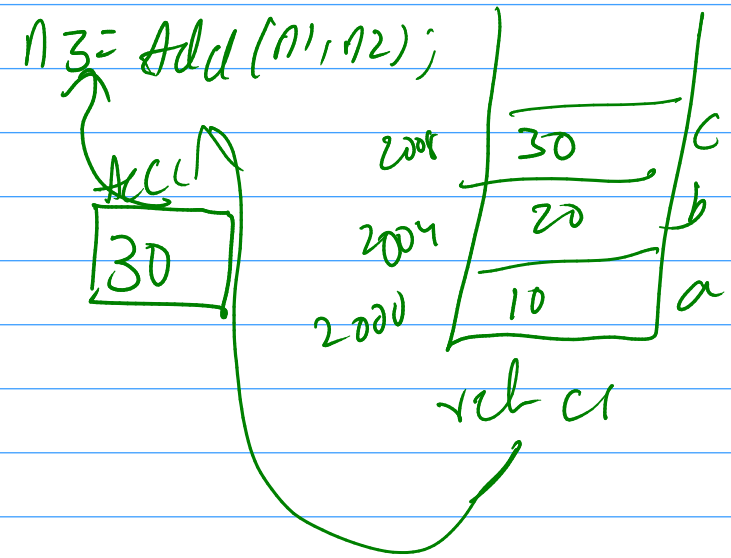
int main(){
int n1=10,n2=20;
Swap(n1,n2);
}
```



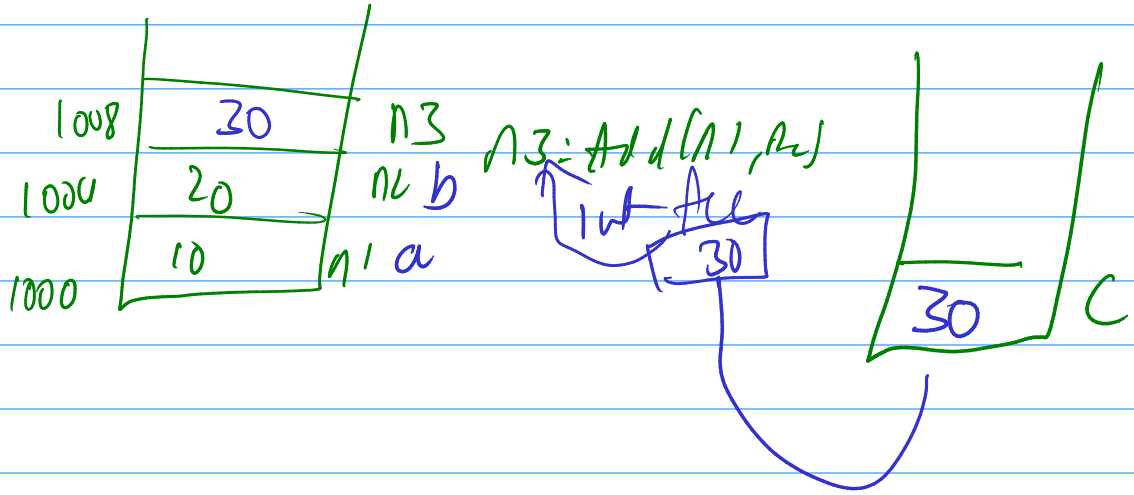
```
int add(int a,int b)
{
int c;
c=a+b;
}

int main()
{ int n1=10,n2=20,n3;
n3=Add(n1,n2);
}
```






```
int Add(int &a,int &b)
{
        int c=a+b;
        return c;
}
```




```
int & Add(int &a,int &b)
{
static int c=a+b;
return c;
}
```

